

2019 Ph H2 Q4

Section: Our Dynamic Universe

Topic: Gravitation — Satellites and field strength

A communications satellite orbits at 36.0×10^6 m above Earth's surface. Mass of Earth = 6.0×10^{24} kg, radius of Earth = 6.4×10^6 m. (a) Calculate distance from Earth's centre. (b) Given gravitational force = 57 N, find satellite mass. (c) Calculate Earth's g at satellite. (d) Second satellite: quarter mass, half the distance — compare forces.

Worked solution

(a) Distance $r = R_E + h = 6.4 \times 10^6 + 36.0 \times 10^6 = 4.24 \times 10^7$ m.

Answer: 4.24×10^7 m

(b) From Newton's law: $F = GMm/r^2$.

$$m = Fr^2/(GM) =$$

$$57 \times (4.24 \times 10^7)^2 / (6.67 \times 10^{-11} \times 6.0 \times 10^{24}).$$

$$= 2.56 \times 10^3 \text{ kg}.$$

Answer: 2.6×10^3 kg

$$(c) g = GM/r^2 = (6.67 \times 10^{-11} \times 6.0 \times 10^{24}) / (4.24 \times 10^7)^2.$$

$$= 0.22 \text{ N kg}^{-1}.$$

Answer: 0.22 N kg^{-1}

(d) Force $\propto m/r^2$.

For 2nd satellite: mass = $\frac{1}{4}m$, distance = $\frac{1}{2}r$.

So ratio = $(\frac{1}{4})/(\frac{1}{2}^2) = 1$.

The gravitational force is the same as before.

Final answers

(a) $r = 4.24 \times 10^7 \text{ m}$

(b) $m \approx 2.6 \times 10^3 \text{ kg}$

(c) $g \approx 0.22 \text{ N kg}^{-1}$

(d) Force is the same as for the first satellite

Revision tips

- Distance from centre = radius + altitude.
- Newton's law: $F = GMm/r^2$.
- Field strength: $g = GM/r^2$.
- If both mass and distance change, compare using proportionality $F \propto m/r^2$.
- Always check units: g in N kg^{-1} , F in N .